UNIVERSITI TUNKU ABDUL RAHMAN
LEE KONG CHIAN FACULTY OF ENGINEERING AND SCIENCE
LABORATORY SAFETY RULES

The following rules must be obeyed by all students in the Computer laboratories of the faculty. Willful or repeated inadvertent noncompliance may result in dismissal or suspension from the laboratories.

I. No entry without permission:

i) Outsiders are not allowed to enter the laboratory without permission.

ii) No student is allowed to enter the laboratory unless permission has been given by a laboratory staff or a lecturer.

II. At work in the laboratory:

i) The computer facilities at the laboratory are to be used for learning purposes only.

ii) Wear your I.D. cards at all times in a proper visible position whenever you come into the computer laboratory.

iii) It is compulsory to follow dress code of UTAR
   a) Covered shoes (no slippers or flip flops)
   b) Long pants and tidy attire

iv) Do not eat or drink in the laboratory.

v) Do not tamper with any computers, printers, or any other associated equipment. If there are problems or faults, report immediately to the staff at the laboratory.

vi) Do not remove any equipment or document from the computer laboratories.

vii) Do not alter the default hardware and software setup of the computers and printers in the computer laboratories.

viii) Do not copy, rename, change, examine or delete files or information belonging to any other user of the University.

ix) Do not install any unauthorized or unlicensed software into the PC of computer laboratories.

x) Do not make any noise, chit-chat, hold group discussions or disturb other users at the laboratory in any way.

xi) Silent your hand phone when you are in the computer laboratories.

xii) Any equipment broken or damaged must be reported immediately to the laboratory staff.

III. Before leaving the laboratory:

i) No student is allowed to take away any equipment, tools or other property of the laboratory without permission.

ii) Please check and remember to take all your personal belongings such as handphone, wallet, pendrive, hard disk, book, bottle, sweeter, and etc.

iii) Lab staffs are not responsible for lost of personal belongings.

Universiti Tunku Abdul Rahman

Faculty	**Lee Kong Chian Faculty of Engineering and Science**
Department:	**Department of Electrical and Electronics Engineering**
Unit Code and Name	**UEEN 1043 Object-Oriented Concepts and Programming Techniques**
Experiment No.:	**6 / 7**
Title of Experiment:	**Data Analytics using Pandas**
Laboratory Room No. and Name:	**Computer lab**
Experiment Duration (hour):	**3 hours**
Number of Student per Group	**1 student**

Objectives

- To practice how to import data from Excel workbook and perform analytic on the data using Pandas module in Python.

Equipment and Materials

Item Description	***Item category**	Quantity estimation (e.g. per set/group of student)
Personal computer	E	1 per student
Python interpreter (Anaconda Spyder)	S	1 per student

***Item category**	
SP	Sample or specimen
C	Consumable
CH	Chemical
W	Labware, glassware, tool, and components
E	Equipment
S	Software

Caution Statement (Optional)

FACULTY OF ENGINEERING AND SCIENCE
UNIVERSITI TUNKU ABDUL RAHMAN

UEEN 1043 Object-Oriented Concepts and Programming Techniques

Practical 6: Data Analytics using Pandas

Objectives

- To practice how to import data from Excel workbook and perform analytic on the data using Pandas module in Python.

Equipment

- Personal computer with a Python interpreter (Anaconda Spyder)

Introduction

Launch Anaconda Spyder IDE for Python and write codes for the following. It is recommended to launch Spyder from within Anaconda Prompt since a lot of modules will be required for pandas.

Exercise 1

Download the Csv file 'ds_salaries.csv' and follow the example given in the lecture.

Exercise 2

Download the Excel file 'A6-merge.xlsx' and read data from different sheets in Excel.

```
import pandas as pd
pd.options.display.max_rows = 5
xlsx = pd.ExcelFile( 'A6-merge.xlsx' )
```

To find out the data type of xlsx and what are the existing sheets in the Excel file.

```
print( type(xlsx) )
print( xlsx.sheet_names )
```

To read all the sheets into a list, with different rows to skip for each excel sheet:

```
sr = [1, 2]
sheets = []
for i, s in enumerate( xlsx.sheet_names ):
    print( s )
    sheets += [ xlsx.parse(s, skiprows=sr[i]) ]
```

To show all the sheets read into Python workspace:

```
print( sheets )
```

As some of the item names have white spaces before the string, to tidy up:

```
for s in sheets:
    s['item name'] = s['item name'].apply( str.strip )
print( sheets )
```

To extract rows for 'apple', create a mask and apply the mask to produce another data frame which is a subset of the original data frame:

```
mask = ( sheets[0]['item name']=='apple' )
df_wm = sheets[0][ mask ]
print( df_wm )
```

Exercise 3 Concatenate Two Data Frames

Write the Python code to combine the transactions for '**apple**' in the purchase and sales excel sheets (A6-merge.xlsx) using **pd.concat** with '**ignore_index=True**' and '**sort=False**' and then store the result in a new data frame with all the rows sorted by the column '**date/time**'. Your result may look like below.

	date/time	item name	cost per unit	units purchased	price per unit	units sold
0	2017-01-02 11:23:25	apple	5.0	20.0	NaN	NaN
1	2017-01-06 15:45:02	apple	5.0	10.0	NaN	NaN
93	2017-01-06 16:37:43	apple	NaN	NaN	15.0	4.0
94	2017-01-07 19:37:50	apple	NaN	NaN	23.0	5.0
95	2017-01-07 19:40:26	apple	NaN	NaN	21.0	3.0
2	2017-01-08 20:55:51	apple	13.0	6.0	NaN	NaN
..	...	...	...	...	...	...
92	2018-06-08 01:21:48	apple	12.0	20.0	NaN	NaN
272	2018-06-10 03:52:59	apple	NaN	NaN	29.0	5.0
273	2018-06-13 05:27:02	apple	NaN	NaN	18.0	5.0
274	2018-06-14 07:07:13	apple	NaN	NaN	33.0	2.0
275	2018-06-16 08:51:27	apple	NaN	NaN	19.0	1.0
276	2018-06-17 09:31:52	apple	NaN	NaN	12.0	3.0